Lecture 24

Lorentz groups and finite-dimensional representations

Objectives. *We build the double cover* $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ *from Hermitian* 2×2 *matrices. We then classify the finite-dimensional irreducible continuous complex representations of both groups by complexifying the real Lie algebra.*

By the end of this lecture you will be able to:

- Prove that $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a surjective covering homomorphism with kernel $\{\pm I\}$.
- Identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.
- Label the irreducible finite-dimensional continuous complex representations of $\mathrm{SL}(2,\mathbb{C})$ by (j_L, j_R) and decide when they descend to $\mathrm{SO}^+(1,3)$.
- Recognise Weyl spinors and four-vectors in this list.

In special relativity an event has coordinates $x = (x_0, \vec{x})$, with time x_0 and position $\vec{x} \in \mathbb{R}^3$. We use units in which light has speed 1. Inertial observers are related by linear maps that preserve the Minkowski form below. Relativistic fields take values in finite-dimensional representations of the Lorentz group or its double cover $\mathrm{SL}(2,\mathbb{C})$.

24.1 Minkowski space and the Lorentz group

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$. The *Minkowski form* is

$$\langle x, y\rangle = x_0 y_0 - \vec{x}\cdot\vec{y} = x^{T}\eta y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

Definition 24.1. The *Lorentz group* is $\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{T}\eta\Lambda = \eta\}$. The *future cone* is $T^+ = \{x \in \mathbb{R}^4 : x_0 > |\vec{x}|\}$.

Let $\Lambda \in \mathrm{O}(1,3)$. Then $\det\Lambda = \pm 1$ and $\Lambda^{-1} = \eta\Lambda^{T}\eta$, so $(\Lambda^{-1})_{00} = \Lambda_{00}$. Since $\langle \Lambda e_0, \Lambda e_0\rangle = 1$,

$$\Lambda_{00}^2 = 1 + \Lambda_{10}^2 + \Lambda_{20}^2 + \Lambda_{30}^2 \geq 1.$$

So either $\Lambda_{00} \geq 1$ or $\Lambda_{00} \leq -1$.

Lemma 24.2. *Let* $\Lambda \in \mathrm{O}(1,3)$. *Then* $\Lambda_{00} \geq 1$ *if and only if* $\Lambda T^+ \subseteq T^+$. *Consequently the elements with* $\Lambda_{00} \geq 1$ *form a subgroup* $\mathrm{O}^+(1,3)$.

Proof. Suppose $\Lambda_{00} \geq 1$ and let $x \in T^+$. Put $w = \Lambda^{-1} e_0$. Then $\langle w, w \rangle = 1$ and $w_0 = (\Lambda^{-1})_{00} = \Lambda_{00} \geq 1$, so $w_0 > |\vec{w}|$. Since Λ preserves the form,

$$(\Lambda x)_0 = \langle e_0, \Lambda x \rangle = \langle w, x \rangle = w_0 x_0 - \vec{w} \cdot \vec{x} \geq w_0 x_0 - |\vec{w}|\,|\vec{x}| > 0.$$

The last step uses $w_0 > |\vec{w}|$ and $x_0 > |\vec{x}|$. Also $\langle \Lambda x, \Lambda x \rangle = \langle x, x \rangle > 0$. These two facts give $\Lambda x \in T^+$. Conversely, if $\Lambda T^+ \subseteq T^+$, then $\Lambda e_0 \in T^+$. So $\Lambda_{00} > 0$, hence $\Lambda_{00} \geq 1$. The cone condition is preserved by products, and $(\Lambda^{-1})_{00} = \Lambda_{00}$. So these elements form a subgroup. ∎

Definition 24.3. The *proper orthochronous Lorentz group* is

$$\mathrm{SO}^+(1,3) = \{\Lambda \in \mathrm{O}(1,3) : \det \Lambda = 1,\ \Lambda_{00} \geq 1\}.$$

By the cone lemma, Lemma 24.2, it is a group. It is closed in $\mathrm{GL}(4,\mathbb{R})$, hence a Lie group by the closed-subgroup theorem (Lecture 3). By the two dichotomies above, for $\det \Lambda$ and for Λ_{00}, it is also open in $\mathrm{O}(1,3)$. So it has the Lie algebra of $\mathrm{O}(1,3)$:

$$\mathfrak{so}(1,3) = \{Y \in \mathfrak{gl}(4,\mathbb{R}) : Y^T \eta + \eta Y = 0\}.$$

The map $Y \mapsto \eta Y$ is a bijection onto the antisymmetric matrices, so $\dim \mathfrak{so}(1,3) = 6$.

A basis consists of the *rotation generators* J_k and the *boost generators* K_k, for $k = 1, 2, 3$. Here $J_k e_0 = 0$, and J_k acts on $\vec{x}$ by $e_k \times \vec{x}$. Moreover $K_k e_0 = e_k$ and $K_k e_k = e_0$, and K_k maps the other two basis vectors to zero. For example,

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $R_3(\theta) = \exp(\theta J_3)$ rotates the (x_1, x_2)-plane by the positive angle θ. The *boost* $B_3(t) = \exp(t K_3)$ fixes x_1, x_2 and acts by

$$x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3).$$

Parity $\mathrm{diag}(1,-1,-1,-1)$ and time reversal $\mathrm{diag}(-1,1,1,1)$ lie in $\mathrm{O}(1,3)$ but outside the open subgroup $\mathrm{SO}^+(1,3)$. The Lie algebra $\mathfrak{so}(1,3)$ carries no information about these elements. We study only $\mathrm{SO}^+(1,3)$.

24.2 The double cover

Let $\sigma_0 = I$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These form a real basis of the space $\mathcal{H}$ of Hermitian 2×2 matrices. For $x \in \mathbb{R}^4$ put

$$X(x) = \sum_{\mu=0}^{3} x_\mu \sigma_\mu = \begin{pmatrix} x_0 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & x_0 - x_3 \end{pmatrix}.$$

Direct expansion gives

$$\det X(x) = \langle x, x\rangle, \qquad x_0 = \tfrac{1}{2}\operatorname{tr} X(x).$$

For $A \in \mathrm{SL}(2,\mathbb{C})$ the matrix $AX(x)A^*$ is again Hermitian. Define $\varphi(A) \in \mathrm{GL}(4,\mathbb{R})$ by

$$X(\varphi(A)x) = A\,X(x)\,A^*. \tag{24.1}$$

The entries of $\varphi(A)$ are real polynomials in the real and imaginary parts of the entries of A. Hence φ is smooth.

Example 24.4 (Diagonal elements)**.** Let $A = \operatorname{diag}(a, a^{-1})$ with $a \in \mathbb{C}^\times$. Then

$$AX(x)A^* = \begin{pmatrix} |a|^2(x_0+x_3) & (a/\bar a)(x_1 - ix_2) \\ (\bar a/a)(x_1+ix_2) & |a|^{-2}(x_0-x_3)\end{pmatrix}.$$

For $a = e^{-i\theta/2}$, the coordinates x_0, x_3 are fixed and $x_1 + ix_2 \mapsto e^{i\theta}(x_1+ix_2)$. For $a = e^{t/2}$, we get $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$, with x_1, x_2 fixed. Put $X_k = -\frac{i}{2}\sigma_k$. Then

$$\varphi(\exp \theta X_3) = R_3(\theta), \qquad \varphi(\exp tiX_3) = B_3(t).$$

Differentiating at 0 gives $d\varphi(X_3) = J_3$ and $d\varphi(iX_3) = K_3$. Also $\operatorname{diag}(e^{i\theta}, e^{-i\theta}) = \exp(-2\theta X_3)$ maps to $R_3(-2\theta)$, as in the convention for $\mathrm{SU}(2) \to \mathrm{SO}(3)$ in Lecture 4. Finally, $\exp(2\pi X_3) = -I$ maps to $R_3(2\pi) = I$.

Let $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ denote $\mathfrak{sl}(2,\mathbb{C})$ regarded as a six-dimensional real Lie algebra.

Theorem 24.5. *The map φ is a surjective homomorphism $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its differential $d\varphi : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$ is an isomorphism of real Lie algebras, and φ is a two-sheeted covering map.*

Proof. Step 1: a homomorphism into $\mathrm{O}^+(1,3)$. Since $(AB)X(AB)^* = A(BXB^*)A^*$, we have $\varphi(AB) = \varphi(A)\varphi(B)$. Since $\det A = 1$, $\det(AXA^*) = \det X$. So $\varphi(A)$ preserves $\langle x, x\rangle$. By polarisation it preserves $\langle x, y\rangle$, so $\varphi(A) \in \mathrm{O}(1,3)$. Its 00 entry is the time component of AIA^*:

$$\varphi(A)_{00} = \tfrac{1}{2}\operatorname{tr}(AA^*) = \tfrac{1}{2}\textstyle\sum_{i,j}|A_{ij}|^2 > 0.$$

By the dichotomy above, $\varphi(A)_{00} \geq 1$.

Step 2: determinant one. We show that $\mathrm{SL}(2,\mathbb{C})$ is path connected. Let $A \in \mathrm{SL}(2,\mathbb{C})$. Let $P = (A^*A)^{1/2}$ be the positive definite square root, and $U = AP^{-1}$. Then $U^*U = I$ and $\det P = |\det A| = 1$, so $U \in \mathrm{SU}(2)$. Diagonalise U and P by the spectral theorem. Since both have determinant 1, we can write $U = \exp Y_1$ and $P = \exp Y_2$ with Y_1 skew-Hermitian, Y_2 Hermitian, and both traceless. Then $t \mapsto \exp(tY_1)\exp(tY_2)$, for $t \in [0,1]$, is a path in $\mathrm{SL}(2,\mathbb{C})$ from I to A. Along it, $\det\varphi$ is continuous with values in $\{\pm 1\}$, and it starts at 1. So $\det\varphi(A) = 1$.

Step 3: the kernel. Suppose $\varphi(A) = I$. Then $AXA^* = X$ for all $X \in \mathcal{H}$. Taking $X = I$ gives $AA^* = I$, so $AXA^{-1} = X$ for all $X \in \mathcal{H}$. Every complex matrix is $X_1 + iX_2$ with $X_1, X_2 \in \mathcal{H}$. So A commutes with all of $M_2(\mathbb{C})$, hence $A = cI$. Then $\det A = 1$ gives $c = \pm 1$. Conversely, $\pm I$ lie in the kernel.

Step 4: surjectivity. This is the main step. Let $\Lambda \in \mathrm{SO}^+(1,3)$ and $v = \Lambda e_0$. First we remove the boost part. The matrix $V = X(v)$ satisfies

$$\det V = \langle v, v\rangle = 1, \qquad \operatorname{tr} V = 2\Lambda_{00} > 0.$$

Its real eigenvalues have product 1 and positive sum, so both are positive. Thus $V = W\operatorname{diag}(\lambda, \lambda^{-1})W^*$ with W unitary and $\lambda > 0$. Put $P = W\operatorname{diag}(\lambda^{1/2}, \lambda^{-1/2})W^*$. Then $P \in \mathrm{SL}(2,\mathbb{C})$, $P^* = P$ and $P^2 = V$. Hence $X(\varphi(P)e_0) = PIP^* = V$, that is, $\varphi(P)e_0 = v$.

Next we treat the rotation part. Put

$$\Lambda' = \varphi(P)^{-1}\Lambda.$$

This element lies in $\mathrm{SO}^+(1,3)$ and fixes e_0. So it preserves $e_0^{\perp} = \operatorname{span}(e_1, e_2, e_3)$, where $-\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Hence $\Lambda' = \operatorname{diag}(1, R)$ with $R \in \mathrm{SO}(3)$. Let $n = Re_3$, a unit vector, and set $N = \sum_k n_k\sigma_k$. This Hermitian matrix satisfies

$$\operatorname{tr} N = 0, \qquad \det N = -|n|^2 = -1,$$

so the eigenvalues of N are ± 1. By the spectral theorem, $N = W'\sigma_3 W'^*$ with W' unitary. We correct the determinant of W' by replacing W' with

$$W'\operatorname{diag}(1, \overline{\det W'}).$$

Diagonal matrices commute with σ_3, so this replacement preserves the diagonalisation of N. The corrected W' belongs to $\mathrm{SU}(2)$.

Then $\varphi(W')e_0 = e_0$, since $W'W'^* = I$. Also $\varphi(W')e_3 = n$, since $W'\sigma_3 W'^* = N$. So $\varphi(W')^{-1}\Lambda'$ fixes e_0 and e_3. It acts on $\operatorname{span}(e_1, e_2)$ by an element of $\mathrm{SO}(2)$, so it equals $R_3(\theta)$ for some θ. By the rotation formula of Example 24.4,

$$\Lambda = \varphi(P)\,\varphi(W')\,R_3(\theta) = \varphi\big(PW'\exp(\theta X_3)\big).$$

In particular, φ maps $\mathrm{SU}(2)$ onto the rotations $\operatorname{diag}(1, R)$.

Step 5: the differential and the covering. Differentiate (24.1) along $\exp(tY)$, using $\exp(tY)^* = \exp(tY^*)$:

$$X(d\varphi(Y)x) = YX(x) + X(x)Y^*, \qquad Y \in \mathfrak{sl}(2,\mathbb{C}). \tag{24.2}$$

Suppose $d\varphi(Y) = 0$. Taking $X = I$ gives $Y^* = -Y$. Then $YX - XY = 0$ for all $X \in \mathcal{H}$, so Y is scalar, as in Step 3. Since $\operatorname{tr} Y = 0$, we get $Y = 0$. Thus $d\varphi$ is injective. Both algebras have real dimension 6, so $d\varphi$ is a linear bijection. The differential of a Lie group homomorphism preserves brackets, so $d\varphi$ is an isomorphism of real Lie algebras.

By Steps 2 and 4, $\mathrm{SO}^+(1,3) = \varphi(\mathrm{SL}(2,\mathbb{C}))$ is connected. A continuous surjective homomorphism of connected Lie groups with bijective differential is a covering map (Lecture 4). Its fibres are cosets of $\ker\varphi = \{\pm I\}$, so it has two sheets. ∎

Formula (24.2) gives $d\varphi(X_k) = J_k$ and $d\varphi(iX_k) = K_k$ for every k. The case $k = 3$ is the diagonal computation of Example 24.4, Exercise 24.1 treats $k = 1$, and the case $k = 2$ is identical. The brackets $[X_i, X_j] = \epsilon_{ijk}X_k$ then transfer through $d\varphi$:

$$[J_i, J_j] = \epsilon_{ijk}J_k, \qquad [J_i, K_j] = \epsilon_{ijk}K_k, \qquad [K_i, K_j] = -\epsilon_{ijk}J_k.$$

24.3 Complexifying the Lorentz algebra

By the double-cover theorem, Theorem 24.5, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$. A complex representation of this real algebra need not be complex linear. For example, $Y \mapsto \bar{Y}$ is a real Lie algebra automorphism that is not complex linear. Complexification treats both kinds together.

Let $\mathfrak{g}$ be a real Lie algebra and V a finite-dimensional complex space. A real Lie algebra map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ extends uniquely to a complex-linear Lie algebra map $\rho_{\mathbb{C}}(Y \otimes z) = z\rho(Y)$ on $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. A complex subspace is ρ-invariant if and only if it is $\rho_{\mathbb{C}}$-invariant. A linear map intertwines two such maps ρ if and only if it intertwines their extensions.

Proposition 24.6. *The complex-linear map*

$$\iota : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}), \qquad Y \otimes z \mapsto (zY,\, z\bar{Y}),$$

is an isomorphism of complex Lie algebras.

Proof. The pair $(zY, z\bar{Y})$ is real bilinear in (Y, z), so ι is well defined and complex linear. Complex conjugation of matrices preserves brackets, so

$$\iota([Y \otimes z, Y' \otimes z']) = \big(zz'[Y,Y'],\, zz'[\bar{Y},\bar{Y}']\big) = [\iota(Y \otimes z), \iota(Y' \otimes z')].$$

Both Lie algebras have complex dimension 6, so it suffices to prove that ι is injective. A general element $Y \otimes 1 + Y' \otimes i$ has image $(Y + iY',\, \bar{Y} + i\bar{Y}')$. If this is zero, then $Y = -iY'$ and $\bar{Y} = -i\bar{Y}'$. Conjugating the first equation gives $\bar{Y} = i\bar{Y}'$. Hence $\bar{Y}' = 0$, so $Y' = 0$ and $Y = 0$. ∎

Since $\overline{iX_k} = -i\bar{X}_k$, we have $\iota(X_k \otimes 1) = (X_k, \bar{X}_k)$ and $\iota(iX_k \otimes i) = (-X_k, \bar{X}_k)$. Hence the elements

$$X_k^L = \tfrac{1}{2}(X_k \otimes 1 - iX_k \otimes i), \qquad X_k^R = \tfrac{1}{2}(X_k \otimes 1 + iX_k \otimes i)$$

satisfy $\iota(X_k^L) = (X_k, 0)$ and $\iota(X_k^R) = (0, \bar{X}_k)$. We identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$ through the complex-linear extension of $d\varphi$. Then $X_k^L = \frac{1}{2}(J_k - iK_k)$ and $X_k^R = \frac{1}{2}(J_k + iK_k)$. These elements satisfy

$$[X_i^L, X_j^L] = \epsilon_{ijk} X_k^L, \qquad [X_i^R, X_j^R] = \epsilon_{ijk} X_k^R, \qquad [X_i^L, X_j^R] = 0.$$

Moreover $J_k = X_k^L + X_k^R$ and $K_k = i(X_k^L - X_k^R)$.

24.4 Classification of finite-dimensional representations

For $m \geq 0$ let $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$, and let $S^m(A)$ be the restriction of $A^{\otimes m}$ to V_m. The entries of $S^m(A)$ are polynomials in the entries of A, so dS^m is complex linear on $\mathfrak{sl}(2,\mathbb{C})$. From the Lie algebra background, dS^m is the irreducible module of highest weight m. The element $H = \mathrm{diag}(1,-1)$ acts with eigenvalues $m, m-2, \dots, -m$. On $\mathrm{SU}(2)$, V_m is the spin-$m/2$ representation.

Definition 24.7. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, the representation (j_L, j_R) of $\mathrm{SL}(2,\mathbb{C})$ is

$$D^{(j_L,j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A}) \quad \text{on} \quad V_{2j_L} \otimes V_{2j_R}.$$

Since $A \mapsto \bar{A}$ is a continuous automorphism of $\mathrm{SL}(2,\mathbb{C})$, this is a continuous representation. Its dimension is $(2j_L+1)(2j_R+1)$.

A representation Π of $\mathrm{SL}(2,\mathbb{C})$ *descends* to $\mathrm{SO}^+(1,3)$ if there is a continuous representation Π' of $\mathrm{SO}^+(1,3)$ with

$$\Pi = \Pi' \circ \varphi.$$

This is the descent of Lecture 4, now through the covering φ.

Theorem 24.8. *(a) Each (j_L, j_R) is irreducible, and distinct pairs give inequivalent representations. Every irreducible finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is equivalent to some (j_L, j_R).*

(b) $D^{(j_L,j_R)}(-I) = (-1)^{2j_L+2j_R}$. Hence (j_L, j_R) descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L + 2j_R$ is even. Up to equivalence, these descended representations are all the irreducible finite-dimensional continuous complex representations of $\mathrm{SO}^+(1,3)$.

(c) The restriction of (j_L, j_R) to $\mathrm{SU}(2)$ is equivalent to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, |j_L - j_R| + 1, \dots, j_L + j_R$.

Proof. (a) Write $D = D^{(j_L,j_R)}$, $m = 2j_L$ and $n = 2j_R$. Since $\overline{\exp tY} = \exp t\bar{Y}$, differentiation gives

$$dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y}), \qquad Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}.$$

So the complex-linear extension is

$$dD_{\mathbb{C}}(Y \otimes z) = dS^m(zY) \otimes 1 + 1 \otimes dS^n(z\bar{Y}) = (dS^m \boxtimes dS^n)\big(\iota(Y \otimes z)\big).$$

Thus, through ι, $dD_{\mathbb{C}}$ is the module $V_m \boxtimes V_n$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. We import from the Lie algebra background: these modules are irreducible and pairwise non-isomorphic, since their highest weights (m,n) differ. Every finite-dimensional irreducible module of $\mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C})$ is one of them.

Now we pass from algebra to group. Every continuous finite-dimensional representation Π is smooth (Lecture 3), with $\Pi(\exp Y) = \exp d\Pi(Y)$ (Lecture 2). By Step 2 of the proof of Theorem 24.5, $\mathrm{SL}(2,\mathbb{C})$ is connected, so it is generated by $\exp \mathfrak{sl}(2,\mathbb{C})$ (Lecture 2). Hence a subspace is Π-invariant if and only if it is $d\Pi$-invariant. A linear map intertwines two group representations if and only if it intertwines their differentials. By the extension facts in § 24.3, both statements pass to $d\Pi_{\mathbb{C}}$. So D is irreducible, and distinct pairs are inequivalent.

Finally, let Π be an irreducible finite-dimensional continuous complex representation. Then $d\Pi_{\mathbb{C}} \circ \iota^{-1} \cong V_m \boxtimes V_n$ for some (m,n). This is the complexified differential of $D^{(m/2,n/2)}$, so $\Pi \cong D^{(m/2,n/2)}$.

(b) We have $S^m(-I) = (-1)^m$ and $\overline{-I} = -I$, so $D(-I) = (-1)^{m+n}$. Covering maps are open. So by the double-cover theorem, Theorem 24.5, φ identifies $\mathrm{SO}^+(1,3)$ with $\mathrm{SL}(2,\mathbb{C})/\{\pm I\}$ as topological groups. By Lecture 4, continuous representations of $\mathrm{SO}^+(1,3)$ correspond to continuous representations of $\mathrm{SL}(2,\mathbb{C})$ that are trivial on $-I$. The correspondence preserves irreducibility and equivalence. Now combine this with (a).

(c) Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. For $U \in \mathrm{SU}(2)$ one has $\bar{U} = \varepsilon U \varepsilon^{-1}$ (Exercise 24.3). So $S^n(\bar{U}) = S^n(\varepsilon)S^n(U)S^n(\varepsilon)^{-1}$, and the restriction is equivalent to $V_m \otimes V_n$. The Clebsch–Gordan rule from the background gives the decomposition. ∎

Every finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is a direct sum of representations (j_L, j_R). This follows from Weyl's complete reducibility theorem for the semisimple algebra $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, transferred as in the proof.

Only $(0,0)$ admits an invariant inner product. The reason is that a boost generator has the nonzero real eigenvalue $j_L + j_R$ on every $(j_L, j_R) \neq (0,0)$. Exercise 24.6 gives the proof.

In physics, let a field ψ take values in a representation space V with matrices $D(A)$. Its transformation is

$$\psi'(x) = D(A)\psi(\varphi(A)^{-1}x).$$

The label (j_L, j_R) classifies the components of ψ. It does not describe the unitary Hilbert space of particle states, which is the subject of Lecture 25.

24.5 Weyl spinors and four-vectors

Example 24.9 (Weyl spinors). The representation $(\frac{1}{2}, 0)$ is $\mathbb{C}^2$ with $A \mapsto A$. The representation $(0, \frac{1}{2})$ is $\mathbb{C}^2$ with $A \mapsto \bar{A}$. Physicists call their vectors left-handed and right-handed Weyl spinors. The choice of hands is a convention. By the classification theorem, Theorem 24.8, the two representations are inequivalent. A direct check uses traces. For $A = \mathrm{diag}(2i, -\frac{i}{2}) \in \mathrm{SL}(2,\mathbb{C})$,

$$\operatorname{tr} A = \tfrac{3i}{2}, \qquad \operatorname{tr} \bar{A} = -\tfrac{3i}{2}.$$

Equivalent representations have equal traces. These traces differ, so the trace check confirms that the two representations are inequivalent. Both representations restrict to V_1 on $\mathrm{SU}(2)$. Both send $-I$ to -1, so neither descends to $\mathrm{SO}^+(1,3)$. The Dirac representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ also sends $-I$ to -1, so it does not descend either.

Example 24.10 (Four-vectors). Let $\Psi : \mathbb{C}^2 \otimes \mathbb{C}^2 \to M_2(\mathbb{C})$ be the linear isomorphism given by $\Psi(\xi \otimes \zeta) = \xi\zeta^T$. The map Ψ sends $e_i \otimes e_j$ to the matrix unit E_{ij}. For $A \in \mathrm{SL}(2,\mathbb{C})$,

$$\Psi(A\xi \otimes \bar{A}\zeta) = A\xi(\bar{A}\zeta)^T = A\xi\zeta^T\bar{A}^T = A\,\Psi(\xi \otimes \zeta)\,A^*.$$

So $D^{(\frac{1}{2},\frac{1}{2})}(A)$ becomes $Z \mapsto AZA^*$ on $M_2(\mathbb{C}) = \mathcal{H} \oplus i\mathcal{H}$. This action preserves $\mathcal{H}$ and acts there by $\varphi(A)$. Hence $(\frac{1}{2}, \frac{1}{2})$ is equivalent to $\mathbb{C}^4 = \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{C}$ with $\varphi(A)$ extended complex-linearly. Since $2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2$ is even, the representation $(\frac{1}{2}, \frac{1}{2})$ descends by the descent criterion of Theorem 24.8. The descended representation is the defining representation of $\mathrm{SO}^+(1,3)$ on $\mathbb{C}^4$.

Two checks are consistent with this identification. First, the restriction rule of Theorem 24.8(c) predicts $V_0 \oplus V_2$ on $\mathrm{SU}(2)$. Indeed $U \in \mathrm{SU}(2)$ acts by $Z \mapsto UZU^{-1}$. It fixes I, which spans the time direction. It preserves the traceless matrices, which are the spatial vectors and carry spin 1. Second, compare boost eigenvalues. In $(\frac{1}{2}, \frac{1}{2})$, the element $iX_3 = \frac{1}{2}\sigma_3$ acts by $\frac{1}{2}(H \otimes 1 + 1 \otimes H)$. Its eigenvalues are $\frac{1}{2}(q + q')$ with $q, q' \in \{\pm 1\}$:

$$1, \quad 0, \quad 0, \quad -1.$$

On $\mathbb{R}^4$, the matrix $d\varphi(iX_3) = K_3$ has eigenvectors $e_0 + e_3$, e_1, e_2, $e_0 - e_3$ with eigenvalues $1, 0, 0, -1$. The lists agree.

The table lists small cases. Exercise 24.4 identifies $(1,0)\oplus(0,1)$ with $\Lambda^2\mathbb{C}^4$, the complexified antisymmetric tensors.

(j_L, j_R)	dimension	on SU(2)	descends	name
$(0,0)$	1	V_0	yes	scalar
$(\frac12,0)$, $(0,\frac12)$	2	V_1	no	Weyl spinors
$(1,0)$, $(0,1)$	3	V_2	yes	summands of $\Lambda^2\mathbb{C}^4$
$(\frac12,\frac12)$	4	$V_0\oplus V_2$	yes	four-vector
$(1,\frac12)$	6	$V_1\oplus V_3$	no	–

24.6 Self-checks

(1) Compute $\varphi(\operatorname{diag}(2,\frac12))$.
(2) Find all $A\in\mathrm{SL}(2,\mathbb{C})$ with $\varphi(A)=R_3(\pi)$.
(3) Does $(\frac32,\frac12)$ descend to $\mathrm{SO}^+(1,3)$? Give its dimension and its restriction to SU(2).

Answer. (1) By the diagonal formula of Example 24.4 with $a=2$, we get $x_0\pm x_3\mapsto 4^{\pm1}(x_0\pm x_3)$. Since $a/\bar a=1$, the coordinates x_1 and x_2 are fixed. This is $B_3(t)$ with $e^t=4$, so $\cosh t=\frac{17}{8}$ and $\sinh t=\frac{15}{8}$. Thus

$$\varphi\big(\operatorname{diag}(2,\tfrac12)\big)=\begin{pmatrix}\frac{17}{8}&0&0&\frac{15}{8}\\0&1&0&0\\0&0&1&0\\\frac{15}{8}&0&0&\frac{17}{8}\end{pmatrix}.$$

Check: $17^2-15^2=64$.

(2) The element $\exp(\pi X_3)=\operatorname{diag}(-i,i)$ maps to $R_3(\pi)$. The fibre is a coset of $\{\pm I\}$, so the answer is $A=\pm\operatorname{diag}(-i,i)$.

(3) Here $2j_L+2j_R=4$ is even, so it descends. Its dimension is $4\cdot2=8$. The restriction is $V_2\oplus V_4$, of dimension $3+5=8$.

24.7 Summary

The action $X\mapsto AXA^*$ on Hermitian matrices preserves $\det X=\langle x,x\rangle$. It gives a two-sheeted covering $\varphi:\mathrm{SL}(2,\mathbb{C})\to\mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. The map ι identifies the complexified Lorentz algebra with $\mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C})$. Up to equivalence, the irreducible finite-dimensional continuous representations of $\mathrm{SL}(2,\mathbb{C})$ are the (j_L,j_R), given by $S^{2j_L}(A)\otimes S^{2j_R}(\bar A)$. Exactly those with $2j_L+2j_R$ even descend to $\mathrm{SO}^+(1,3)$. Weyl spinors are $(\frac12,0)$ and $(0,\frac12)$, and four-vectors are $(\frac12,\frac12)$.

24.8 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001, Chapter II, Section 3, on finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$.